6 Discussion

[To be completed: Discussion outline. (1) How much of measured connectedness is a common factor? About 41% of Diebold–Yilmaz connectedness disappears without the factor (range 38–43% across robustness checks); one factor explains 58% of the variance. What does this say about contagion versus common exposure (Forbes and Rigobon, 2002; Rey, 2013)? Why do the common share and DY connectedness move together, and why is that partly mechanical? (2) Does the method change the picture? Lasso and DY almost coincide, so regularization is not what matters; NETS keeps the pattern at a lower level; FNETS measures a different object (idiosyncratic part). Which method would you use for which purpose (risk monitoring, contagion analysis, portfolio diversification)? (3) Transmitters and receivers: South Africa, Korea, Taiwan versus Turkey, Vietnam, Argentina; how much may reflect trading hours of US-listed funds; what changes once the factor is removed. (4) Limitations: returns instead of volatilities; ETF prices embed exchange rates and US trading hours; short windows for NETS and FNETS (positive-definite adjustment, empty idiosyncratic network in the full sample, tuning-driven drops); almost no lagged links survive in the sparse methods.]

7 Conclusion

[To be completed: Conclusion outline. Answer the research question in three or four sentences. What can be claimed with confidence: weekly connectedness among emerging markets is high and mostly contemporaneous; a large and stable share of it (about 40%) is a common factor; the dense methods agree with each other. What cannot be claimed: the exact direction of spillovers between countries, or the idiosyncratic network structure, given the short windows. One sentence on extensions (volatility connectedness, longer samples, more markets).]

8 References

Matteo Barigozzi and Christian Brownlees. NETS: Network estimation for time series. *Journal of Applied Econometrics*, 34(3):347–364, 2019.

Matteo Barigozzi, Haeran Cho, and Dom Owens. FNETS: Factor-adjusted network estimation and forecasting for high-dimensional time series. *Journal of Business & Economic Statistics*, 42(3):890–902, 2024.

T. Tony Cai, Weidong Liu, and Harrison H. Zhou. Estimating sparse precision matrix: Optimal rates of convergence and adaptive estimation. *The Annals of Statistics*, 44(2):455–488, 2016.

Jiahua Chen and Zehua Chen. Extended Bayesian information criteria for model selection with large model spaces. *Biometrika*, 95(3):759–771, 2008.

Francis X. Diebold and Kamil Yılmaz. On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics*, 182(1):119–134, 2014.

Kristin J. Forbes and Roberto Rigobon. No contagion, only interdependence: Measuring stock market comovements. *The Journal of Finance*, 57(5):2223–2261, 2002.

Mario Forni, Marc Hallin, Marco Lippi, and Lucrezia Reichlin. The generalized dynamic-factor model: Identification and estimation. *Review of Economics and Statistics*, 82(4):540–554, 2000.

Marc Hallin and Roman Liška. Determining the number of factors in the general dynamic factor model. *Journal of the American Statistical Association*, 102(478):603–617, 2007.

Gary Koop, M. Hashem Pesaran, and Simon M. Potter. Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74(1):119–147, 1996.

Dom Owens, Haeran Cho, and Matteo Barigozzi. fnets: An R package for network estimation and forecasting via factor-adjusted VAR modelling, 2024. R package version 0.1.6, CRAN.

Jie Peng, Pei Wang, Nengfeng Zhou, and Ji Zhu. Partial correlation estimation by joint sparse regression models. *Journal of the American Statistical Association*, 104(486):735–746, 2009.

H. Hashem Pesaran and Yongcheol Shin. Generalized impulse response analysis in linear multivariate models. *Economics Letters*, 58(1):17–29, 1998.

Hélène Rey. Dilemma not trilemma: The global financial cycle and monetary policy independence. In *Proceedings of the Economic Policy Symposium, Jackson Hole*, pages 285–333. Federal Reserve Bank of Kansas City, 2013.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1):267–288, 1996.

Hui Zou. The adaptive lasso and its oracle properties. *Journal of the American Statistical Association*, 101(476):1418–1429, 2006.

9 Data and Code Availability

The data (weekly ETF returns) and the full code are available at `https://github.com/ezellabra/ML4E-code`. The pipeline is written in Python; NETS and FNETS are estimated in R (packages `nets` 0.9.1 and `fnets` 0.1.6) through a command-line bridge. All parameters are in `config.yaml` and the whole analysis is reproduced with `make all` (or `python run_all.py`).

A Robustness

Table 3: Robustness: mean total connectedness (%) by method, share of DY connectedness lost once the common factor is removed (FNETS), and FNETS common-variance share.

Scenario	Windows	DY	Lasso VAR	NETS	FNETS (idio.)	Lost w/o factor (%)	Common var. (%)
Baseline (w=104, H=4)	179	81.0	80.0	61.5	47.2	41	58
H=2	179	80.9	80.0	61.5	47.2	41	58
H=8	179	81.0	80.1	61.5	47.2	41	58
w=78	185	81.2	79.6	57.8	50.5	38	58
w=130	172	81.1	80.5	63.4	46.2	43	58
Excluding SPY	179	80.0	78.9	59.9	45.9	42	58
Excluding windows touching 2020	141	79.7	78.6	58.8	47.2	41	55

Table 3 reruns every method under alternative choices. The levels of Diebold–Yilmaz and lasso connectedness hardly move: between 79.7% and 81.2% and between 78.6% and 80.5% across all scenarios. NETS is the most sensitive to the window length (57.8% with 78-week windows, 63.4% with 130-week windows), as shorter windows leave the penalized estimator less information and make it keep fewer links. FNETS idiosyncratic connectedness ranges from 45.9% to 50.5%. The horizon matters very little: Diebold–Yilmaz changes by about 0.1 points between $H = 2$ and $H = 8$, and the sparse methods not at all, because their estimated lag coefficients are almost all zero and their variance decompositions are essentially driven by the contemporaneous covariance. Excluding the S&P 500 lowers Diebold–Yilmaz connectedness by one point, and removing every window that contains a week of 2020 (141 windows left) lowers Diebold–Yilmaz, lasso and NETS connectedness by one to three points and leaves FNETS unchanged, without changing the ranking of the methods.